PRACTICAL 4

1.Adam is working in an IT company. He has been given a task to reduce the load of a system by killing some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation?

Theory:

This practical focuses on understanding process management in the Linux operating system. It involves in the Linux studying how processes are created, identified, executed & terminated using system calls & commands. The experiment demonstrates the use of process identifies such as PID & PPID, creation of parent & child processes using fork () system call and processes like orphan & zombies.

```
m309@m309-BY-OEM:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 15:52 ?        00:00:06 /sbin/init splash
root           2       0  0 15:52 ?        00:00:00 [kthreadd]
root           3       2  0 15:52 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 15:52 ?        00:00:00 [kworker/R-rcu_g]
root           5       2  0 15:52 ?        00:00:00 [kworker/R-rcu_p]
root           6       2  0 15:52 ?        00:00:00 [kworker/R-slub_]
root           7       2  0 15:52 ?        00:00:00 [kworker/R-netns]
root           9       2  0 15:52 ?        00:00:00 [kworker/0:0H-events_highpri]
root          10       2  0 15:52 ?        00:00:00 [kworker/0:1-events]
root          12       2  0 15:52 ?        00:00:00 [kworker/R-mm_pe]
root          13       2  0 15:52 ?        00:00:00 [rcu_tasks_kthread]
root          14       2  0 15:52 ?        00:00:00 [rcu_tasks_rude_kthread]
root          15       2  0 15:52 ?        00:00:00 [rcu_tasks_trace_kthread]
root          16       2  0 15:52 ?        00:00:00 [ksoftirqd/0]
root          17       2  0 15:52 ?        00:00:00 [rcu_preempt]
root          18       2  0 15:52 ?        00:00:00 [migration/0]
root          19       2  0 15:52 ?        00:00:00 [idle_inject/0]
root          20       2  0 15:52 ?        00:00:00 [cpuhp/0]
root          21       2  0 15:52 ?        00:00:00 [cpuhp/1]
root          22       2  0 15:52 ?        00:00:00 [idle_inject/1]
root          23       2  0 15:52 ?        00:00:00 [migration/1]
root          24       2  0 15:52 ?        00:00:00 [ksoftirqd/1]
root          26       2  0 15:52 ?        00:00:00 [kworker/1:0H-events_highpri]
root          27       2  0 15:52 ?        00:00:00 [cpuhp/2]
root          28       2  0 15:52 ?        00:00:00 [idle_inject/2]
root          29       2  0 15:52 ?        00:00:00 [migration/2]
```

- Kill processes by name

```
m309@m309-BY-OEM:~$ kill 9367
bash: kill: (9367) - No such process
m309@m309-BY-OEM:~$ pkill firefox
m309@m309-BY-OEM:~$ nano fork_demo.c
m309@m309-BY-OEM:~$ gcc fork_demo.c -o fork_demo
m309@m309-BY-OEM:~$ ./fork_demo
Parent Process
PID: 9442
Child PID: 9443
Child Process
PID: 9443
PPID: 9442
m309@m309-BY-OEM:~$
```

Kill a process based on the process name

```
m309        9361    9354  0 17:01 pts/0    00:00:00 bash
m309        9374    8413  0 17:02 ?        00:00:00 /snap/chromium/3293/usr/lib/chromium-browser/chrome --typ
root        9388       2  0 17:02 ?        00:00:00 [kworker/3:2]
root        9389       2  0 17:02 ?        00:00:00 [kworker/1:0-cgroup_destroy]
m309        9391    9361  0 17:02 pts/0    00:00:00 ps -ef
m309@m309-BY-OEM:~$ ps -ef |grep firefox
m309        9393    9361  0 17:02 pts/0    00:00:00 grep --color=auto firefox
m309@m309-BY-OEM:~$ ps -ef | grep firefox
m309        9407    9361  0 17:03 pts/0    00:00:00 grep --color=auto firefox
m309@m309-BY-OEM:~$ kill 9407
bash: kill: (9407) - No such process
```

· Kill a single process at a time with the given process ID

```
m309@m309-BY-OEM:~$ kill 9407
bash: kill: (9407) - No such process
m309@m309-BY-OEM:~$ pkill firefox
```

2. Write a program for process creation using C

Orphan Process

- Orphan Process:

→ An orphan process is a child process whose parent process terminates before the child finishes execution. The orphan process is adopted by the init or system process

Code-

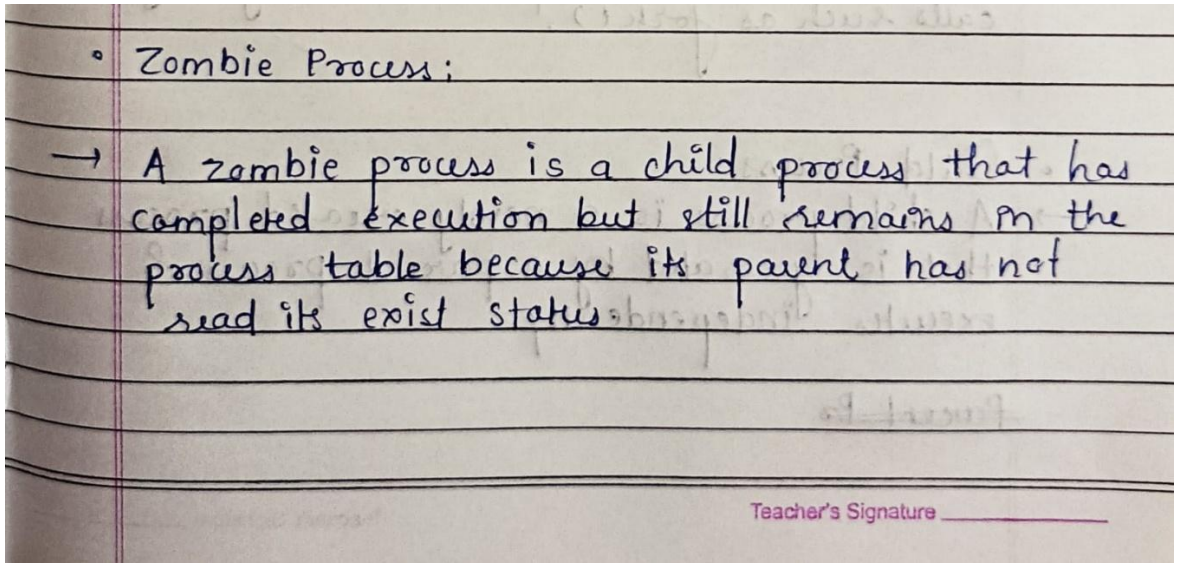
```
GNU nano 7.2                                                    orphan.c
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid > 0) {
        // Parent process
        printf("Parent process exiting\n");
    } else {
        // Child process
        sleep(5);
        printf("Child process running\n");
    }
    return 0;
}
```

```
m309@m309-BY-OEM:~$ nano orphan.c
m309@m309-BY-OEM:~$ gcc orphan.c -o orphan
m309@m309-BY-OEM:~$ ./orphan
Parent exiting
m309@m309-BY-OEM:~$ Child PID: 9506
New Parent PID: 6892
```

- Zombie Process



• Zombie Process:

→ A zombie process is a child process that has completed execution but still remains in the process table because its parent has not read its exist status.

Teacher's Signature _____

```
GNU nano 7.2                                        zombie .c
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        // Child process
        printf("Child exiting\n");
        return 0;
    } else {
        // Parent process
        printf("Parent sleeping, child becomes zombie\n");
        sleep(30);   // VERY IMPORTANT
    }
    return 0;
}
```

```
nano zombie.c
m309@m309-BY-OEM:~$ gcc zombie.c -o zombie
m309@m309-BY-OEM:~$ ./zombie
Child process exiting
```

3. Create the process using fork () system call.

- Child Process creation
- Parent process creation
- PPID and PID

- Process:
A process is an instance of a program that is currently being executed in the operating system. It includes program code, data, stack & system resources.

- Process ID (PID):

→ Process ID (PID) is a unique numerical identifier assigned by the operating system to each running process for identification & management

- Parent Process

→ A parent process is a process that creates one or more child processes using system calls such as fork().

- Child Process:
→ A child process is a newly created process that is generated by a parent process & executes independently.

~~Parent Pr~~

Teacher's Signature

```
m309@m309-BY-OEM:~$ nano fork_demo.c
```

```
GNU nano 7.2                                    fork_demo.c
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
    } else {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
    }
    return 0;
}
```

4.Infinite Loop

```
m309@m309-BY-OEM:~$ nano loop.c
m309@m309-BY-OEM:~$ gcc loop.c -o loop
m309@m309-BY-OEM:~$ ./loop
Running...
Running...
Running...
Running...
Running...
Running...
```

```
GNU nano 7.2                                    loop.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    while (1) {
        printf("Running...\n");
        sleep(1);
    }
    return 0;
}
```